



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°8_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

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L'objectif de ce TD est de:

1. Mettre en place une solution complète de supervision d'un serveur Ubuntu :

- collecte des métriques système
- visualisation graphique
- détection d'incidents
- alertes

Mettons en place une solution complète permettant de surveiller :CPU, RAM, espace disque, trafic réseau, service web Nginx

Installation de Prometheus

```
lens@lens:~$ sudo useradd --no-create-home --shell /bin/false prometheus
lens@lens:~$ sudo mkdir /etc/prometheus
lens@lens:~$ sudo mkdir /var/lib/prometheus
```

Téléchargement de Prometheus

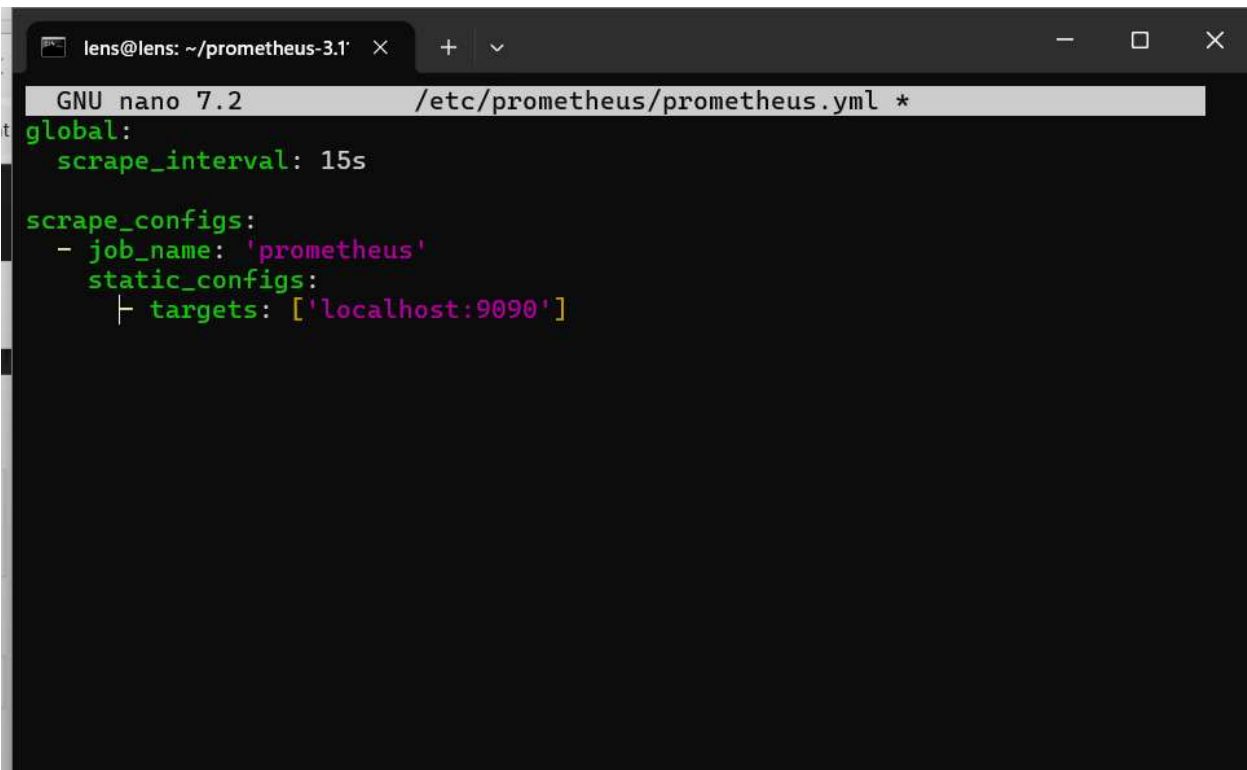
```
[Lens@Lens:~]$ wget https://github.com/prometheus/prometheus/releases/download/v3.11.0/prometheus-3.11.0.linux-amd64.tar.gz
--2026-04-26 19:49:21-- https://github.com/prometheus/prometheus/releases/download/v3.11.0/prometheus-3.11.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 148.82.112.3
Connecting to github.com (github.com)[148.82.112.3]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /releases-assets.githubusercontent.com/github-production-release-asset/6838921/87173d66-03b4-43e7-be3b-621bc478f08c?sp=https&se=2026-04-26T20%3A26%3AO7Zrscsd=attachment%3B%fileaname%3Dprometheus-3.11.0.linux-amd64.tar.gz&nset=application%2Foctet-stream&a=dd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-26T19%3A26%3AO4Z&se=2026-04-26T20%3A26%3AO7Z&sk=b8UbyAYxSRlPyuf8nALGQIve4ZzsOCjddv4GA%3D&jwt=eYJ0eXAoI0JKV1QiLCJhbGciOiJIUzE1NiJ9.eYJpc3MiOiJnaXNodWYuY29tIiwiaXYkYXIvcmljaWVuLnCmVudHVsZWZfc2ZSlnBlbnRlbmcuY29tIiwia2VSIjoia2VMSIsImVucCmVCMTC3MzInJlZjUwMawbmJmJmxoxNCzMjMyOTYyLClCmVXRoiJCioamVsZWZfc2ZWFzc2V0ChVJVHVhdGVmbGlvdjV1bG9mlmVudhvmHAFAQRORLHOH6nAB8flhdtPwZEor3lpuzG8&response-content-disposition=attachment%3B%20filename%3Dprometheus-3.11.0.linux-amd64.tar.gz&riication%2Foctet-stream [following]
--2026-04-26 19:49:24-- https://releases-assets.githubusercontent.com/github-production-release-asset/6838921/87173d66-03b4-43e7-be3b-018-11-89&r=b&sp=https&se=2026-04-26T20%3A26%3AO7Zrscsd=attachment%3B%fileaname%3Dprometheus-3.11.0.linux-amd64.tar.gz&nset=application%2Foctet-stream&a=dd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-26T19%3A26%3AO4Z&se=2026-04-26T20%3A1-09&sig=Nv3RL8UBYAYxSRlPyuf8nALGQIve4ZzsOCjddv4GA%3D&jwt=eYJ0eXAoI0JKV1QiLCJhbGciOiJIUzE1NiJ9.eYJpc3MiOiJnaXNodWYuY29tIiwiaXYkYXIvcmljaWVuLnCmVudHVsZWZfc2ZWFzc2V0ChVJVHVhdGVmbGlvdjV1bG9mlmVudhvmHAFAQRORLHOH6nAB8flhdtPwZEor3lpuzG8&response-content-disposition=attachment%3B%20filename%3Dprometheus-3.11.0.linux-amd64.tar.gz&riication%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.110.133, 185.199.109.133, 185.199.199.199, 185.199.199.199
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.110.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 150167744 (143M) [application/octet-stream]
Saving to: 'prometheus-3.11.0.linux-amd64.tar.gz.2'
```

```
lens@lens:~$ gzip -t prometheus-3.11.0.linux-amd64.tar.gz
lens@lens:~$ tar -xvzf prometheus-3.11.0.linux-amd64.tar.gz
prometheus-3.11.0.linux-amd64/
prometheus-3.11.0.linux-amd64/prometheus
prometheus-3.11.0.linux-amd64/LICENSE
prometheus-3.11.0.linux-amd64/prometheus.yml
prometheus-3.11.0.linux-amd64/NOTICE
prometheus-3.11.0.linux-amd64/promtool
lens@lens:~$ cd prometheus-3.11.0.linux-amd64
lens@lens:~/prometheus-3.11.0.linux-amd64$ |
```

Copie des fichiers

```
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo cp prometheus /usr/local/bin/
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo cp promtool /usr/local/bin/
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo cp prometheus.yml /etc/prometheus/
lens@lens:~/prometheus-3.11.0.linux-amd64$ |
```

Configuration principale



The screenshot shows a terminal window with a nano editor editing the file /etc/prometheus/prometheus.yml. The configuration is as follows:

```
GNU nano 7.2 /etc/prometheus/prometheus.yml *
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']
```

Service system

```
lens@lens: ~/prometheus-3.1' x + v
GNU nano 7.2 /etc/systemd/system/prometheus.service *
[Unit]
Description=Prometheus
After=network.target

[Service]
User=prometheus
ExecStart=/usr/local/bin/prometheus \
--config.file=/etc/prometheus/prometheus.yml \
--storage.tsdb.path=/var/lib/prometheus/

[[Install]]
WantedBy=multi-user.target
```

Donner les permissions

```
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo chown -R prometheus:prometheus /etc/prometheus
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo chown -R prometheus:prometheus /var/lib/prometheus
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo chmod -R 775 /var/lib/prometheus
lens@lens:~/prometheus-3.11.0.linux-amd64$ |
```

Démarrage

```

lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo systemctl daemon-reload
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo systemctl start prometheus
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
lens@lens:~/prometheus-3.11.0.linux-amd64$ sudo systemctl status prometheus
● prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; prese>
   Active: active (running) since Sun 2026-04-26 21:12:33 UTC; 22s ago
 Main PID: 1961 (prometheus)
    Tasks: 6 (limit: 2263)
   Memory: 30.8M (peak: 31.3M)
      CPU: 602ms
   CGroup: /system.slice/prometheus.service
           └─1961 /usr/local/bin/prometheus --config.file=/etc/prometheus>

Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.065Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.067Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.067Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.067Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.067Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.072Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.072Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.074Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.074Z level=>
Apr 26 21:12:34 lens prometheus[1961]: time=2026-04-26T21:12:34.074Z level=>
lines 1-20/20 (END)

```

Interface Web

Prometheus tourne mais mon navigateur n'arrive pas à le rejoindre ce qui empêchera le monitoring avec Grafana et Prometheus.



Hmmm... can't reach this page

192.168.43.79 took too long to respond

Try:

- Search the web for [192 168 43 79](#)
- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_TIMED_OUT

Refresh

Installation de Node Exporter

```

lens@lens:~$ sudo useradd --no-create-home --shell /bin/false node_exporter
lens@lens:~$ wget https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node_exporter-1.10.2.linux-amd64.tar.gz
--2026-04-26 21:50:11-- https://github.com/prometheus/node_exporter/releases/download/v1.10.2/node_exporter-1.10.2.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.112.4
Connecting to github.com (github.com)|140.82.112.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/23923325-96f1-4212-8814-13e7435df395?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-04-26T22%3A37%3A26Z&rscd=attachment%3B+filename%3Dnode\_exporter-1.10.2.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-26T21%3A37%3A24Z&ske=2026-04-26T22%3A37%3A26Z&skvs=b&skv=2018-11-09&sig=d%2Bt0WyqbWQPrJfIMD9%2F%2F7wUE6QhZvH38Be4iCeLiabg%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiOiJoaicmVsZWZzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3NzI0MjAxMiwibmJmIjoxNzc3MjQwMjEyLCJwYXRoIjoicmVsZWZzZWZzc2V0cHJvZHVjdGlubi5ibG9iLmNvcmlUud2luZG93cy5uZXQifQ.b182Ymk4SZ28A\_vxhXkbMORXvD\_MfvTzmoJwbv1IbpQ&response-content-disposition=attachment%3B%20filename%3Dnode\_exporter-1.10.2.linux-amd64.tar.gz

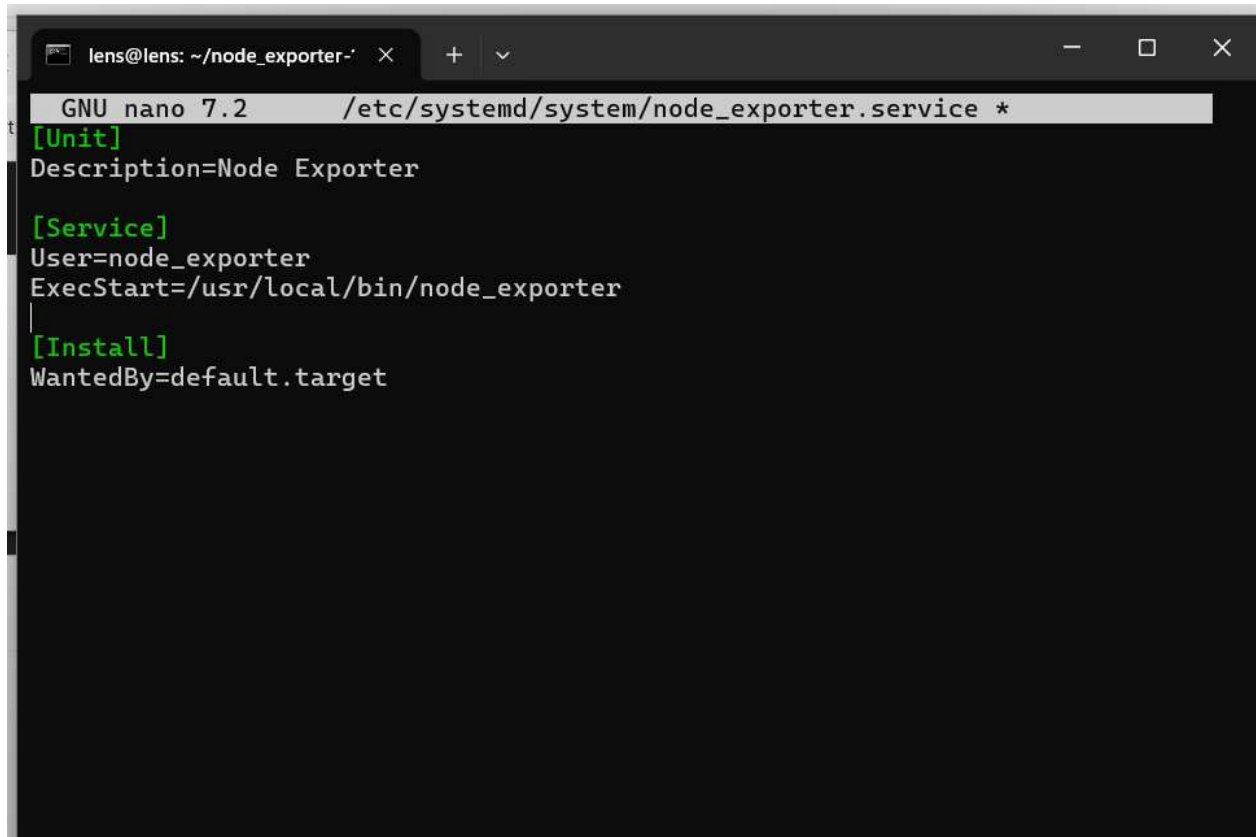
```

```

lens@lens:~$ tar xvf node_exporter-1.10.2.linux-amd64.tar.gz
node_exporter-1.10.2.linux-amd64/
node_exporter-1.10.2.linux-amd64/node_exporter
node_exporter-1.10.2.linux-amd64/LICENSE
node_exporter-1.10.2.linux-amd64/NOTICE
lens@lens:~$ cd node_exporter-1.10.2.linux-amd64
lens@lens:~/node_exporter-1.10.2.linux-amd64$ sudo cp node_exporter /usr/local/bin/
lens@lens:~/node_exporter-1.10.2.linux-amd64$ |

```

Service node_exporter



The image shows a terminal window with a dark background. At the top, a window title bar indicates the user is 'lens' on a host named 'lens', with the current directory being '~/node_exporter-'. Below the title bar, the terminal shows the GNU nano 7.2 editor editing the file '/etc/systemd/system/node_exporter.service'. The file content is as follows:

```
[Unit]
Description=Node Exporter

[Service]
User=node_exporter
ExecStart=/usr/local/bin/node_exporter

[Install]
WantedBy=default.target
```

Démarrage

```

lens@lens:~/node_exporter-1.10.2.linux-amd64$ sudo systemctl daemon-reload
lens@lens:~/node_exporter-1.10.2.linux-amd64$ sudo systemctl start node_exporter
lens@lens:~/node_exporter-1.10.2.linux-amd64$ sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/default.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.service.
lens@lens:~/node_exporter-1.10.2.linux-amd64$ sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset=enabled)
   Active: active (running) since Sun 2026-04-26 21:55:00 UTC; 51s ago
     Main PID: 4544 (node_exporter)
        Tasks: 3 (limit: 2263)
       Memory: 2.0M (peak: 2.3M)
          CPU: 15ms
      CGroup: /system.slice/node_exporter.service
              └─4544 /usr/local/bin/node_exporter

Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.973Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.974Z lev>
Apr 26 21:55:00 lens node_exporter[4544]: time=2026-04-26T21:55:00.974Z lev>
lines 1-20/20 (END)

```

Test

Puisque Prometheus ne marche pas on ne vas assurément pas trouver les metrics

Ajout de Node Exporter à Prometheus

```
lens@lens: ~/node_exporter-  
GNU nano 7.2 /etc/prometheus/prometheus.yml *  
global:  
  scrape_interval: 15s  
  
scrape_configs:  
  - job_name: 'prometheus'  
    static_configs:  
      - targets: ['localhost:9090']  
  
  - job_name: 'node_exporter'  
    static_configs:  
      - targets: ['localhost:9100']
```

Redémarrage

```
lens@lens:~/node_exporter-1.10.2.linux-amd64$ sudo systemctl restart prometheus
```

Installation de Grafana

```
lens@lens:~$ sudo apt install -y software-properties-common apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.49.4).
software-properties-common set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 72 not upgraded.
Need to get 3,970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Get:1 http://ht.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3,970 B]
Fetched 3,970 B in 1s (4,379 B/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 159903 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Setting up apt-transport-https (2.8.3) ...
Scanning processes...
Scanning linux images...
```

```
lens@lens:~$ sudo add-apt-repository "deb https://packages.grafana.com/oss/deb stable main"
Repository: 'deb https://packages.grafana.com/oss/deb stable main'
Description:
Archive for codename: stable components: main
More info: https://packages.grafana.com/oss/deb
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Adding deb entry to /etc/apt/sources.list.d/archive_uri-https_packages_grafana_com_oss_deb-noble.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/archive_uri-https_packages_grafana_com_oss_deb-noble.list
Hit:1 http://ht.archive.ubuntu.com/ubuntu noble InRelease
Get:2 https://packages.grafana.com/oss/deb stable InRelease [7,661 B]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://ht.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://ht.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 https://repo.zabbix.com/zabbix/7.0/ubuntu noble InRelease
Err:2 https://packages.grafana.com/oss/deb stable InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 963FA27710458545
Reading package lists... Done
W: GPG error: https://packages.grafana.com/oss/deb stable InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 963FA27710458545
E: The repository 'https://packages.grafana.com/oss/deb stable InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
```

```

lens@lens:~$ sudo apt install grafana -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  grafana
0 upgraded, 1 newly installed, 0 to remove and 72 not upgraded.
Need to get 274 MB of archives.
After this operation, 1,028 MB of additional disk space will be used.
Get:1 https://packages.grafana.com/oss/deb stable/main amd64 grafana amd64 1
3.0.1 [274 MB]
Fetched 274 MB in 42min 54s (107 kB/s)
Selecting previously unselected package grafana.
(Reading database ... 159907 files and directories currently installed.)
Preparing to unpack .../grafana_13.0.1_amd64.deb ...
Unpacking grafana (13.0.1) ...
Setting up grafana (13.0.1) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `grafana' (UID 116) ...
info: Adding new user `grafana' (UID 116) with group `grafana' ...

lens@lens:~$ sudo systemctl enable --now grafana-server
Synchronizing state of grafana-server.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.s
ervice → /usr/lib/systemd/system/grafana-server.service.
lens@lens:~$

```

- Bien que Grafana et Prometheus soient installés, configurés et actifs, l'accès externe échoue. Les tests démontrent que le problème est lié à la connectivité réseau de la VM. L'IP attribuée n'est pas joignable depuis l'hôte, ce qui empêche l'ouverture des interfaces web. Donc au final je ne peux pas avancer avec le monitoring

Questions

Pourquoi utiliser Prometheus + Grafana plutôt que Zabbix seul ?

On utilise Prometheus avec Grafana plutôt que Zabbix seul parce que Prometheus collecte efficacement les métriques dans des environnements dynamiques et Grafana permet de les visualiser en temps réel avec des tableaux de bord modernes, alors que Zabbix est plus adapté aux infrastructures classiques mais moins flexible pour le cloud.

2. Différence entre Node Exporter et App Exporters ?

Le Node Exporter sert à recueillir les métriques du système comme le CPU, la mémoire ou le disque, tandis que les App Exporters collectent les métriques spécifiques aux applications, par exemple MySQL ou Redis.

3. Pourquoi Grafana peut afficher des métriques en temps réel ?

Grafana peut afficher des métriques en temps réel parce qu'il interroge directement des bases de données de séries temporelles comme Prometheus et actualise les tableaux de bord en continu.

4. Comment ajouter un nouvel hôte à ce monitoring ?

Pour ajouter un nouvel hôte au monitoring, il faut installer un exporter sur la machine et l'ajouter dans la configuration de Prometheus, ou bien déployer un agent Zabbix et enregistrer l'hôte dans l'interface de Zabbix.

Cocclusion.

Ce TD m'a permis de savoir comment mettre en place une solution complète de supervision d'un serveur Ubuntu avec Prometheus et Grafana. donc la tache a été réussie.